MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

1/ TYPE 8964

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic and metal

See figure 1

Mounting position: Any

Weight: 2.5 ounces (71 grams) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	F MHz	Ef V (Note 1)	Eb Vdc	epy kv	Ec Vdc	ec v	Ib mAdc	ib a	ic a	tp µs
Maximum:										
Anode pulsed freq multiplier:	3,000	6.0 ±5%	---	3.5	---	-150	---	3.0	1.5	6.0
TEST CONDITIONS:	---	6.0	1,000	---	Adj	---	100	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit:	Du ---	Pp W	Pg W (Note 2)	tk sec (min)	TE °C (Note 3)	T °C (Anode shank) (Note 3)	Cooling --- (Notes 3 & 5)
Maximum:							
Anode pulsed freq multiplier:	0.0033	100	1.5	60	250	250	---
TEST CONDITIONS:	---	---	---	300	---	---	Note 6

GENERAL:

Qualification - Required

Compatibility testing - Required (see note 7)

1/ Formerly Y-561

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	4							
1301	Heater current	-		0.65	II	If	0.90	1.05	A
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec	-0.5	-5.5	Vdc
1266	Total grid current	-		0.65	II	Ic	---	-8.0	µAdc
1211	Insulation of electrodes	-	Eb = Ek = 0; Ec = -500 Vdc	0.65	II	Rgk	50	---	Meg
1231	Pulsing emission	-	eb = ec = etd/is = 6.0 a; tp = 3 µs (max); prr = 600 (max)	0.65	II	etd	---	200	v
1316	Amplification factor	-	Eb = 500 Vdc; Ib = 50 mAdc	0.65	II	Mu	150	180	---
	Quality conformance inspection, part 2								
1261	Electrode voltage (2) (grid)	8	Ec/Ib = 1.0 mAdc	---	---	Eco	---	-25.0	Vdc
1331	Direct-interelectrode capacitance (grounded cathode connection)	-	Fixture per Drawing 158-JAN	---	---	Cin Cgp Cout	6.30 1.85 ---	7.50 2.15 0.035	pF pF pF
---	Resonance	9	No voltages	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	10	Group C; Ef = 6.0 V; filament standby; t = 500 hours	---	---	---	---	---	---
---	Life-test end point	10		---	---	Δib	---	25	%
1031	Variable-frequency vibration	11,12	F = 55 to 500 to 55 Hz; Accel = 10 G peak (min); Ebb = 400 Vdc; Rp = 10,000 ohms; Ec/Ib = 10 mAdc	---	---	Ep	---	250	mVac
1002	Barometric pressure, reduced	11,14	Pressure = 35 mmHg (max); voltage = 1,800 Vac; TA = 30° ±10°C	---	---	---	---	---	---
---	Torque	11,13	No voltages	---	---	---	---	---	---
1042	Shock, specified pulse	11	Condition A; no voltages; fixture per Drawing 280-JAN	---	---	---	---	---	---
---	Torque and shock test end point:								
1266	Total grid current	-		---	---	Ic	---	-10	µAdc

NOTES:

1. The transit-time heating effect of the cathode may have to be compensated for by a reduction in the heater voltage after dynamic operation of the tube has started. This back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for any particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to 500 MHz.
2. The maximum instantaneous peak grid voltage for pulse service shall be within the range of +250 to -750 volts.

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NOTES: -Continued

3. Sufficient conduction, convection, or forced-air cooling shall be provided for the anode shank and all seals to limit the maximum temperature to the specified value under all operating conditions. Where emphasis is placed on long and reliable life, lower temperatures should be maintained.

4. All tests listed under QCI, part 1, shall be performed at the conclusion of the holding period.

5. With an anode dissipation of 100 watts and with an incoming air temperature of 25°C maximum at sea level a minimum airflow of 12.5 cfm shall be directed across the anode cooler, using the cowl as shown in Drawing 157-JAN.

6. In all electrical tests involving application of heater voltage, conduction, convection, and/or forced-air cooling may be used to assure operation within the maximum rated temperature limits.

7. In addition to the requirements specified herein, compatibility testing shall be required in the AN/URN-3 and AN/GRN-9 equipments. A sample of 20 tubes shall be supplied to Naval Air Test Center (NESTED), Patuxent River, Maryland 20670, on a noninterference basis, to be tested for freedom from oscillation, adequate gain and tuning capability, and good tracking, in both low and high band heads. Also, of the 20 tubes supplied, a sample of four tubes shall be selected for life testing. Each tube shall achieve a minimum of 500 hours operation. In the event of no failures the sample shall be considered as acceptable when a total of 1,600 hours has been accumulated.

8. The electrode voltage (2) grid test may be performed alternately in the following manner:

   Eb = 600 Vdc; Ec/Ib = 1.0 mAdc; and the limit for Eco shall be -15.0 Vdc (max).

9. Grid-anode resonance. Test in cavity per Drawing 278-JAN. Cavity shall resonate at 1,354 ±2.0 MHz with tuning slug per Drawing 277-JAN at TA = 25° ±5°C.

   Grid-cathode resonance. Test in cavity per Drawing 283-JAN. Cavity shall resonate at 1,719 ±2.0 MHz with tuning slug per Drawing 277-JAN at TA = 25° ±5°C.

   When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (MHz)	
	C-gp	C-gk	F-gp	F-gk
1	1.85	6.30	1,965	1,730
2	1.85	6.30	2,015	1,780
3	2.15	7.50	1,850	1,690
4	2.15	7.50	1,900	1,740

10. At zero hours, establish the drive conditions necessary to obtain ib = 2.0a (peak anode current) with Eb = 2,000 Vdc and a bias voltage Ec = -40 Vdc. The pulse width (tp) of the modulator shall be 2 µs minimum and the duty (Du) shall be 0.0025 maximum. With the drive level determined at zero hours, check the anode current at end of life. The maximum allowable drop in anode current (Δib) shall be 25 percent.

11. This test will be performed quarterly, with sampling as follows:

$n_1 = 4$ $c_1 = 0$
$n_2 = 4$ $c_2 = 1$; Where c_2 represents the total allowable failures for the 1st and 2nd samples combined.

All tests shall be considered as nondestructive except in case of failure, and separate samples may be used for each test at the option of the manufacturer. In case of failure after double sampling, the failing test shall become a QCI, part 2, AQL 6.5, inspection level S3 test, for three consecutive successful submissions, at which time the testing may revert to the QCI, part 3 basis.

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NOTES: -Continued.

12. The tube shall be mounted in a socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximate logarithmic progression, and shall require 4 minutes minimum, 6 minutes maximum, to traverse the range. Each tube shall be vibrated for 30 minutes in each position X and Z, except that if the cumulative result of tests on 50 or more tubes of a construction show that more than 75 percent of the tubes have higher output voltages in one position, subsequent measurements need be taken only in the position giving the higher reading. The voltages specified herein shall be applied to the tube during vibration. The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of vibration shall be measured with a suitable device. This device shall have an appropriate voltage range and shall have the ability to measure with an error of less than 10 percent the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibrational output, Ep, shall not exceed the limit specified herein at any point in the sweep-frequency range during the last complete cycle of vibration.

13. The torque test procedure shall be as follows:

    (a) A torque of 15 inch-pounds shall be applied between the anode contact surface and the cathode without perceptible shock.
    (b) A torque of 40 inch-pounds shall be applied between the anode contact surface and the grid without perceptible shock.

14. The applied voltage shall be 60 Hz, between the anode and grid. No other voltages shall be applied. There shall be no evidence of failure as indicated by arc-over.

User activities:
Navy - AS, OS, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

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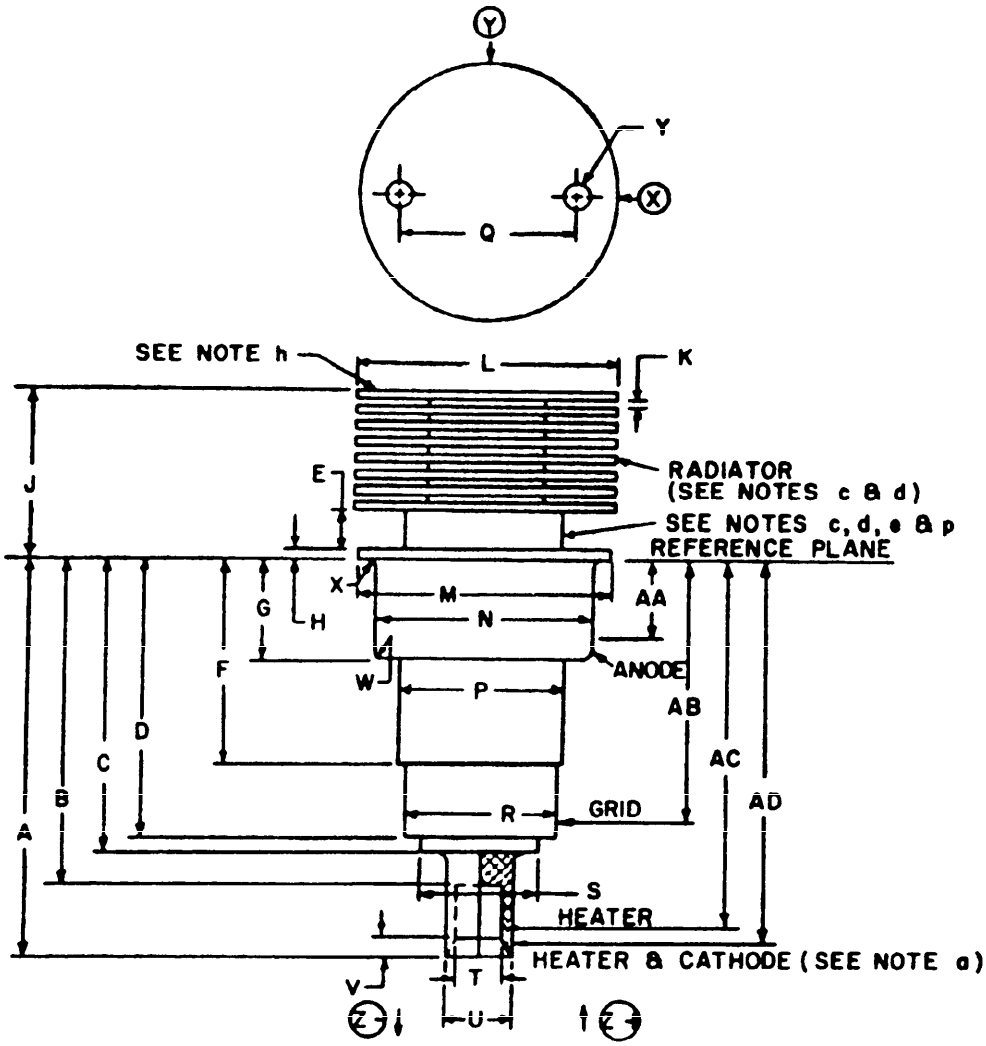


FIGURE 1. Outline drawing of electron tube type 8964.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	1.770 (44.96)	1.830 (46.48)	
B		1.475 (37.47)	
C		1.405 (35.69)	
D	1.217 (30.91)	1.257 (31.93)	
F	0.892 (22.66)	0.932 (23.67)	
G	0.462 (11.73)	0.477 (12.12)	
J	0.766 (19.46)	0.826 (20.98)	
N	1.025 (26.04)	1.035 (26.29)	m, c, k
R	0.655 (16.64)	0.665 (16.89)	m, c, k
T	0.213 (5.41)	0.223 (5.66)	m, c, g, k
U	0.315 (8.00)	0.325 (8.26)	m, c, g, k
Quality conformance inspection, part 3			11
E	0.125 (3.18)	0.185 (4.70)	
H		0.040 (1.02)	
K	0.025 (0.64)	0.046 (1.17)	j
L	1.234 (31.34)	1.264 (32.11)	
M	1.180 (29.97)	1.195 (30.35)	
P	0.752 (19.10)	0.792 (20.12)	n
Q	0.650 (16.51)	0.850 (21.59)	
S		0.545 (13.84)	
V		0.086 (2.18)	
W		0.100 (2.54) RAD	
X		0.035 (0.89) RAD	
Y	0.105 (2.67)	0.145 (3.68)	f
Electrode contact areas			b
AA	0.035 (0.89)	0.361 (9.17)	
AB	1.113 (28.27)	1.193 (30.30)	
AC	1.311 (33.30)	1.505 (38.23)	
AD	1.272 (32.31)	1.612 (40.94)	

NOTES:
a. Insulation material between heater and heater-cathode shall be securely affixed.
b. Dimensions in electrode contact area table are for socket design purposes and are not intended for inspection purposes.
c. Silver plated 30 MSI minimum.
d. Plating not required over radiator and radiator support of copper, aluminum, or approved equivalent.
e. This surface shall be used for measurement of anode shank temperature.
f. Holes for tube extractor through top fin only.
g. Inner edge of heater and outer edge of cathode rf connection shall be free from burrs and sharp edges.
h. This fin shall withstand a 6-inch (152.40 mm) drop test without loosening and without distortion as judged by ability to maintain dimension K.
j. Distortion of fins permissible provided distance between adjacent fins at any point on circumference meets tolerances for dimension K.
k. Eccentricity of contact surfaces shall be gaged from center line of reference shall be as follows:

Contact surface	TIR maximum	Reference
Anode	0.020	Cathode
Grid	0.020	Cathode
Heater	0.012	Cathode

m. Diameters N, R, T, and U shall apply throughout entire contact areas as defined by dimensions AA, AB, AC, and AD respectively.
n. This surface shall not be used for clamping or locating.

FIGURE 1. Outline drawing of electron tube type 8964 - Continued.

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